PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—PRELIM PAPER 1—
MEMO

19 September 2023

SECTION A – Climate and Weather

QUESTION 1: Short Questions

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Terrestrial radiation refers to …

A. An area of a slope that receives permanent sun.
B. Heat given off by Earth.
C. Heat absorbed by Earth.
D. An area of a slope that only receives morning sun. (1)

1.1.2. A south facing slope in South Africa would experience ...

A. Warmer conditions between December and February.
B. Colder conditions between December and February.
C. Consistently warmer conditions throughout the year.
D. Consistently cooler conditions throughout the year. (1)

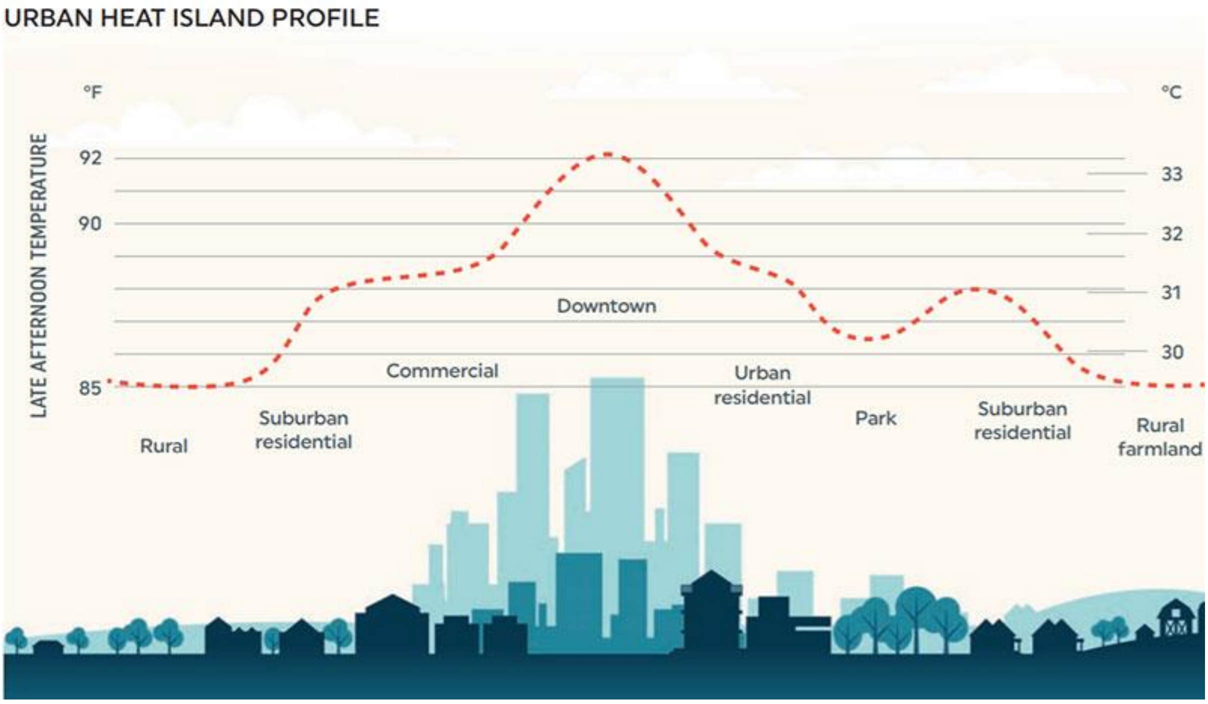
1.1.3. Katabatic winds refer to …

A. Winds that blow up valley sides and along the valley floor during the day.
B. Winds that travel vertically up in cities due to excess heat sources.
C. Sinking winds resulting from temperature inversions.
D. Winds that blow down valley sides and along the valley floor during the night. (1)

1.1.4. Anabatic winds refer to …

A. Winds that blow up valley sides and along the valley floor during the day.
B. Winds that travel vertically up in cities due to excess heat sources.
C. Sinking winds resulting from temperature inversions.
D. Winds that blow down valley sides and along the valley floor during the night. (1)

Refer to the diagram below showing an Urban Heat Island Profile and answer the questions that follow.



1.1.5. An Urban Heat Island can be defined as …

A. A phenomenon whereby the central part of a city is warmer than the surrounding rural areas.

B. A phenomenon whereby the central part of a city is cool than the surrounding rural areas.

C. A phenomenon whereby there is increased pollution in the air that is trapped by an inversion layer.

D. A combination of pollution, sinking inversion layers and intense heat increasing the temperature in the city centre. (1)

1.1.6. A possible reason for the drop in temperature at the park may be …

A. Additional air conditioning units being used to cool the plants.

B. There is more traffic in this area creating more wind.

C. Vast areas of vegetation absorbing heat.

D. There is always more rain over parklands than urban centres. (1)

1.1.7. During which season are we most likely to experience urban heat islands?

A. Autumn & Spring.

B. Winter & Summer.

C. All year.

D. Winter only. (1)

1.1.8. When thinking about precipitation it should be noted that …

A. Rural areas receive more rain that urban areas due to having less heat and more vegetation present.

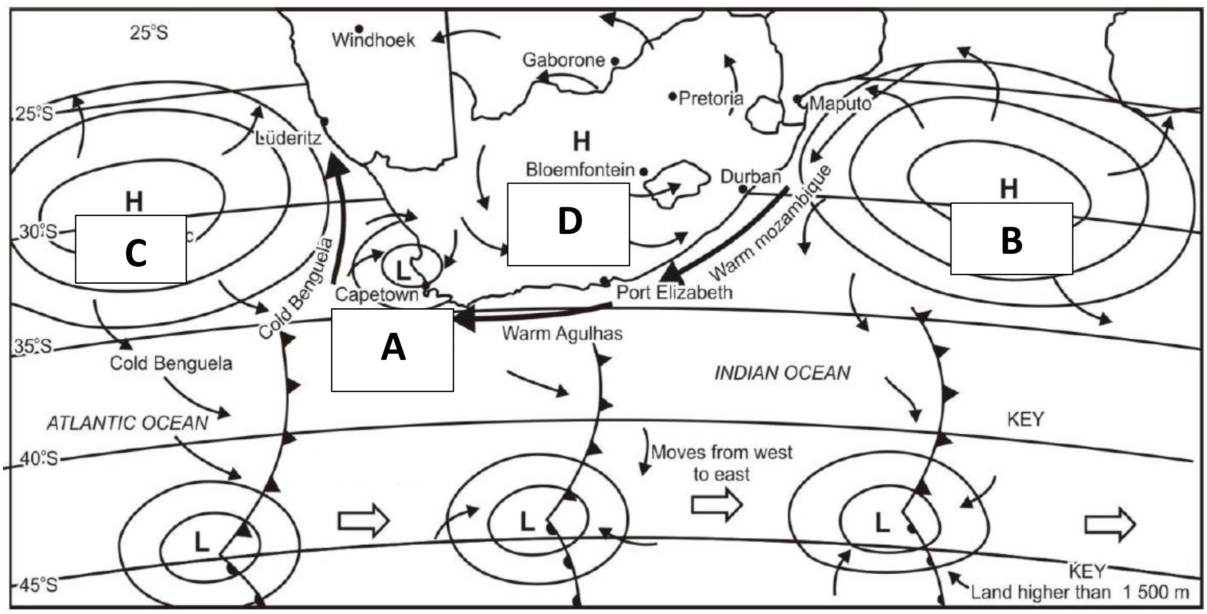
B. Urban areas experience more thunderstorms due to the increased heat and hygroscopic nuclei present.

C. Suburban residential areas receive more rain as they are the perfect middle between cool temperatures and increased vegetation.

D. Downtown will experience less frequent storms due to the high-rise buildings preventing the natural flow of wind. (1)

(8 x 1) (8)

1.2. Refer to the diagram below and answer the questions that follow. For each answer select the correct word in brackets. Write only the question and the correct word. E.g. 1.2.8 Jam.



1.2.1. The low pressure around Cape Town (A) would be an example of a (Berg Wind/Cut off low/Coastal low pressure).
1.2.2. The name of the high pressure cell at B is (South Atlantic/South Indian/Kalahari) Anticyclone.
1.2.3. The wind coming into South Africa from C has a moisture content that is (more/less/the same) as the wind coming from B.
1.2.4. The season depicted in the diagram would likely be (Summer/Winter/Spring).
1.2.5. The air that circulates around the high pressure at D will rotate in a/an (clockwise/anticlockwise/upwards) direction.
1.2.6. The latitude at which the warmer Westerlies meet the cold Polar Easterlies is known as the (Sub-polar collision/Polar Front/Upfront meet).
1.2.7. Bloemfontein is likely experiencing (partly cloudy/overcast/clear) conditions in this diagram.

(7 x 1) (7)

1	**B**	**1**	**Coastal Low**
2	**D**	**2**	**South Indian**
3	**D**	**3**	**Less**
4	**A**	**4**	**Winter**
5	**A**	**5**	**Anticlockwise**
6	**C**	**6**	**Polar Front**
7	**C**	**7**	**Clear**
8	**B**		

[15]

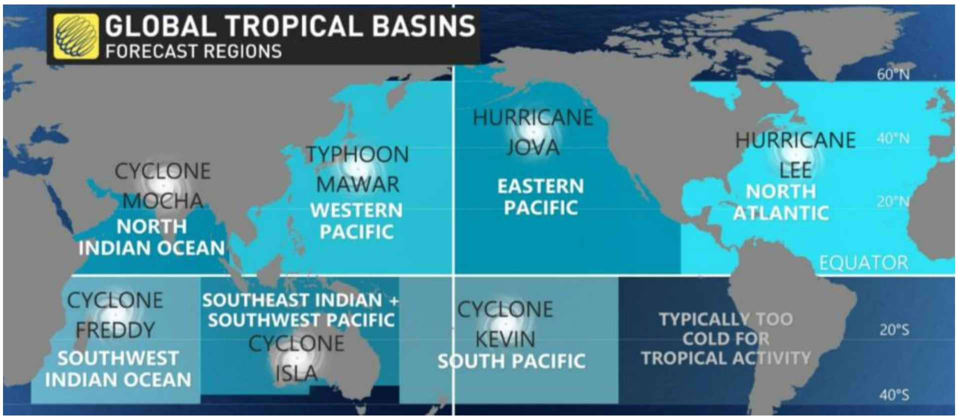
QUESTION 2: Tropical Cyclones

2. Refer to the article below and answer the questions that follow.

A world first, every tropical ocean saw a Category 5 storm in 2023

For the first time since weather satellites began peering at our skies from above, each of the world's warm oceans have seen a scale-topping tropical cyclone so far in 2023.

While we'd typically see at least one tropical system somewhere on Earth achieve this infamous milestone each year, we've never seen each of the world's tropical ocean basins pump out a scale-topping storm all within the same year.



The first Category 5 tropical cyclone of 2023 was a deadly powerhouse of a storm in the Indian Ocean that's likely to secure a spot in record books as the longest-lived tropical cyclone ever observed.

Cyclone Freddy formed off the northwestern coast of Australia on February 6, continuing west across the entire Indian Ocean for more than a month. This storm caused record breaking destruction across the continent of Africa.

Hurricanes Jova and Lee exploded just hours apart

Two hurricanes rapidly intensified on either side of North America over the course of two days in September.

Jova formed far off the western coast of Mexico where it encountered very warm waters over the eastern Pacific Ocean. The system intensified from a tropical storm with 110 km/h winds on September 5, 2023, into a Category 5 hurricane with 260 km/h just 24 hours later.

The following day, Hurricane Lee over in the Atlantic Ocean accomplished a similar feat. Lee rapidly grew its maximum sustained winds from 130 km/h to 270 km/h over a 24-hour period, one of the fastest bouts of intensification ever observed over the Atlantic.

Mawar grew into the year's strongest tropical cyclone

But the year's strongest tropical system developed in the western Pacific Ocean back in May. Super Typhoon Mawar swiftly grew into a monstrous Category 5 equivalent storm when it peaked with maximum winds measuring as high as 295 km/h.

Three more storms round out 2023's historic feat

The hot waters around Oceania gave rise to another Category 5 cyclone at the end of February. Cyclone Kevin's intense winds briefly reached the top of the scale after traversing the islands of Vanuatu.

2.1. How many typhoons occurred before Super Typhoon Mawar developed? (1 x 2) (2)

12

2.2. What is unique about the 'cyclone season' of 2023 according to the article? (1 x 1) (1)

First season to have a category 5 in all oceans

2.3. What do you think may have caused the cyclone season of 2023 to be so unique? (1 x 2) (2)

[Accept logical answers]

Global warming – Warmer oceans across the globe have led to more cyclones developing

2.4. In the southern hemisphere, which section of a tropical cyclone would be the most destructive? The bottom left or the top left quadrant? (1 x 1) (1)

Bottom left

2.5. When considering hurricanes Jova and Lee, quote ONE piece of evidence from the article to support the notion that climate change may be responsible for these storms developing so rapidly. (1 x 1) (1)

"Encountered very warm waters over the eastern Pacific Ocean"

2.6. In a paragraph of approximately EIGHT lines, discuss TWO social impacts of tropical cyclones and TWO strategies we can implement to mitigate the damage caused by these phenomena. (4 x 2) (8)

[Accept logical answers]

Social Impacts – MUST BE A DESCRIPTION

Family losses, job losses, mental health impacts, orphaned children, infrastructure damage (must link to social such as educational institutions destroyed etc.)

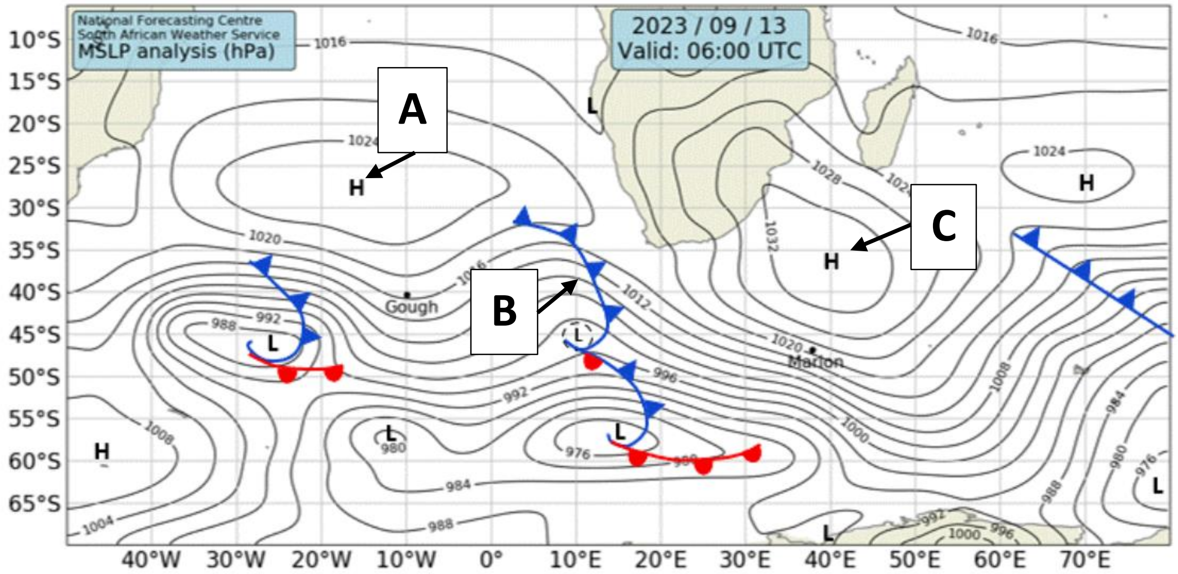
Mitigation strategies – MUST BE A DESCRIPTION

Prediction and warning community, plan in advance, rescue operations in place etc.

[15]

QUESTION 3: Anti-cyclones

3. Study the synoptic weather map below and answer the questions that follow.



3.1. What is the name given to the anticyclone present directly over the east coast of South Africa? (1 x 1) (1)

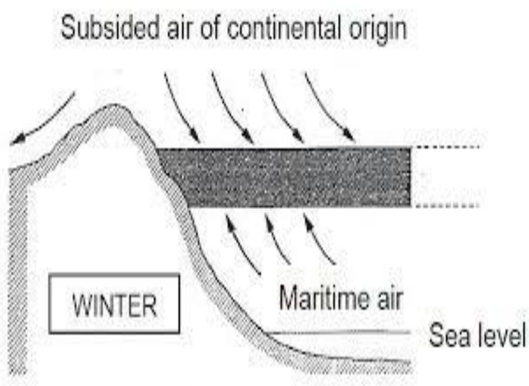
South Indian AC

3.2. Provide labels for the pressure systems labelled A and B on the diagram. (2 x 1) (2)

A – South Atlantic HP

B – Mid-latitude cyclone/cold front

3.3. Draw a simple cross section showing how the high pressure named at C prevents rain from entering the interior? (2 x 2) (4)



2 marks – movement of air (HP air descending)

2 marks – LP/Ocean air not getting past inversion layer

3.4. Over the next few **weeks,** are temperatures likely to increase or decrease across South Africa? (1 x 1) (1)

Increase

3.5. Provide ONE piece of evidence to substantiate your answers to QUESTION 3.4. (1 x 1) (1)

Date (accept season is Spring)

3.6. Considering your answer to QUESTION 3.4, briefly describe what will happen to the Kalahari High Pressure system over the next few weeks. (2 x 2) (4)

[Accept reasonable answers]

Interior will heat up as we approach summer

Rising air over the interior will lift the Kalahari HP

Inversion layer will rise allowing coastal air into the interior

3.7. Describe ONE positive consequence for inland farmers as a result of the change in the Kalahari High Pressure system as described in QUESTION 3.6. (1 x 2) (2)

[Accept logical answers]

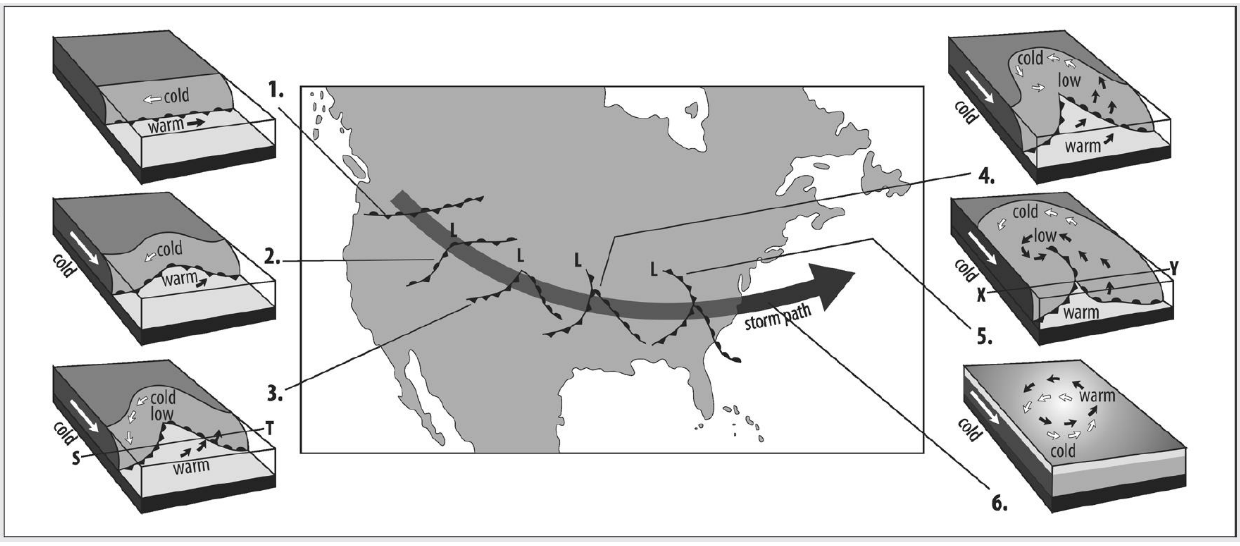
More rain can be expected

Change in temperature means less risk of frost damaging crops

[15]

QUESTION 4: Mid-Latitude Cyclones

4. Study the figure below showing the formation of a mid-latitude cyclone over North America and answer the questions that follow.



4.1. Which hemisphere is this mid-latitude cyclone forming in? (1 x 1) (1)

Northern

4.2. Match the stages (1–6) with the following descriptions (A–F). Write only the stage and the letter that aligns with the stage. E.g. Stage 1 – H

A Cyclone degenerates.

B Occlusion begins.

C Winds begin to spiral.

D Wave forms in polar front.

E Fronts develop.

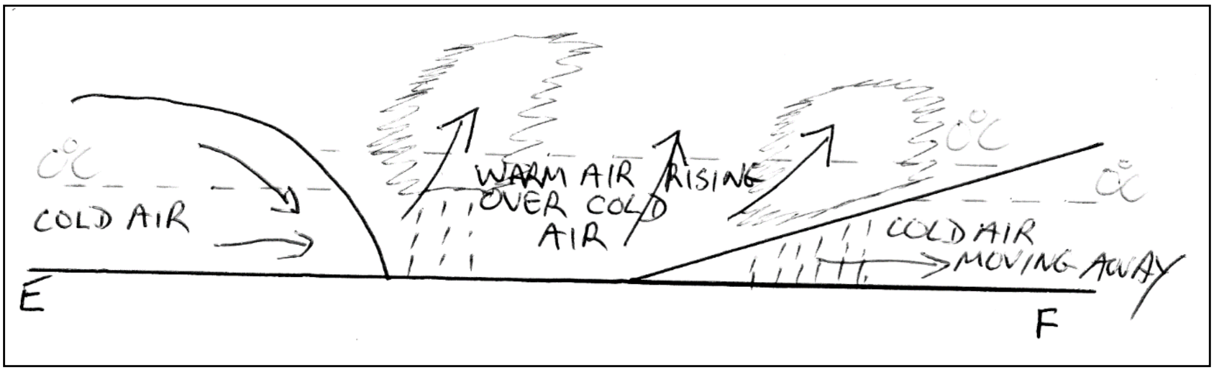
F Fully mature mid-latitude cyclone is formed. (6 x 1) (6)

1	D
2	C
3	E
4	F
5	B
6	A

4.3. Draw a simple cross section view from S to T (no. 3). Include the following:

Front symbols

Temperatures of all air pockets (5 x 1) (5)



Front symbols – 2 marks (cold and warm fronts)

Temperatures of all air pockets – 3 marks (Cold, warm, cold)

4.4. What is the likely cloud type that would develop at the cold front? (1 x 1) (1)

Cumulonimbus

4.5. Explain the term 'occlusion'. (1 x 2) (2)

[Concept]

Both fronts combine to form a single, rainy front.

[15]

TOTAL SECTION A: [60]

SECTION B - Geomorphology

QUESTION 5: Short Questions

5.1. The following descriptions are all features of a river system. State whether each of these is found in the UPPER, MIDDLE or LOWER course of a river. E.g. 5.1.8. Upper

5.1.1. Erosion is dominant at this stage.

5.1.2. In an ideal transverse river profile, erosion and deposition would be equal at this stage.

5.1.3. Rapids and waterfalls are a feature here.

5.1.4. Most headward erosion takes place at this stage.

5.1.5. Oxbow lakes typically form here.

5.1.6. Rivers may meander here, however there is still vertical erosion taking place.

5.1.7. Deltas are a common feature here.

(7 x 1) (7)

1	Upper
2	Middle
3	Upper
4	Upper
5	Lower
6	Middle
7	Lower

5.2. Choose a term from COLUMN B that matches the geomorphological description in COLUMN A. Write only the letter (A–I) next to the question number (5.2.1–5.2.8) in the ANSWER BOOK, for example 5.2.9 J.

COLUMN A	COLUMN B
5.2.1. Stream pattern that flows into a central pan or a low-lying area.	A. Rectangular pattern.
5.2.2. Forms on igneous rocks that have joints and cracks.	B. Superimposed.
5.2.3. Forms on inclined rock layers that are unequally resistant to erosion.	C. Trellis pattern.
	D. Antecedent drainage.
5.2.4. Develops on a dome where streams flow outwards.	E. Radial/centrifugal pattern.
5.2.5. A drainage pattern that is maintained even after the land has been uplifted and/or folded.	F. Centripetal pattern.
5.2.6. A stream pattern that does not match the geology and topography of the existing landscape.	G. Dendritic pattern.
	H. Braided pattern.
5.2.7. Forms in mostly glacial regions where no specific pattern can be identified.	I. Deranged pattern.
5.2.8. Occurs on rocks that have uniform resistance to erosion and where tributaries join the main river at acute angles.	

(8 x 1) (8)

1	F
2	A
3	G
4	E
5	D
6	B
7	I
8	C

[15]

QUESTION 6: Drainage basins

6. Refer to the figure below showing a drainage basin and answer the questions that follow.

6.1. Why do the plants look terrified in the cartoon? (1 x 2) (2)

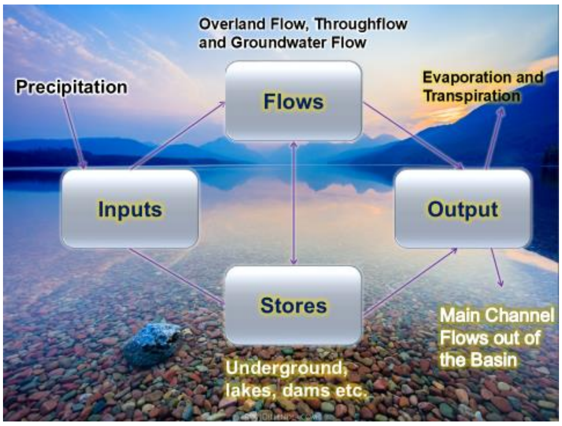
[Accept reasonable answers]

The water is poisoned

6.2. Where is the water coming from in the cartoon? (1 x 1) (1)

The Vaal

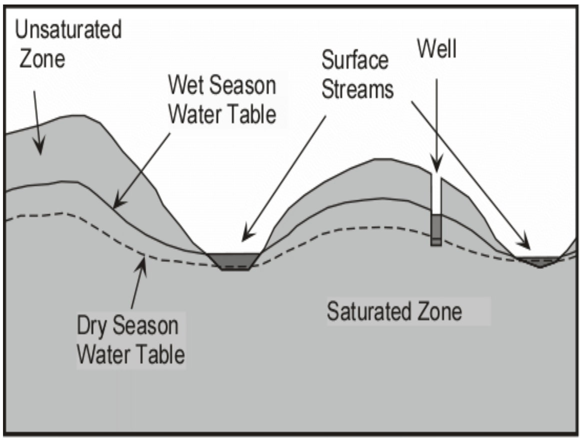
6.3. Draw a simple flow diagram showing how water moves through a drainage basin. NOTE: Your flow diagram should contain the words Input, flows, stores and outputs. (4 x 1) (4)



6.4. The Vaal river is a permanent river. Draw TWO simple diagrams illustrating the difference in water table levels between a permanent and a periodic river. Indicate the following on both diagrams:

a) Dry season water table.

b) Wet season water table. (2 x 2) (4)



6.5. Describe TWO factors that influence the water table levels in any particular area. (2 x 2) (4)

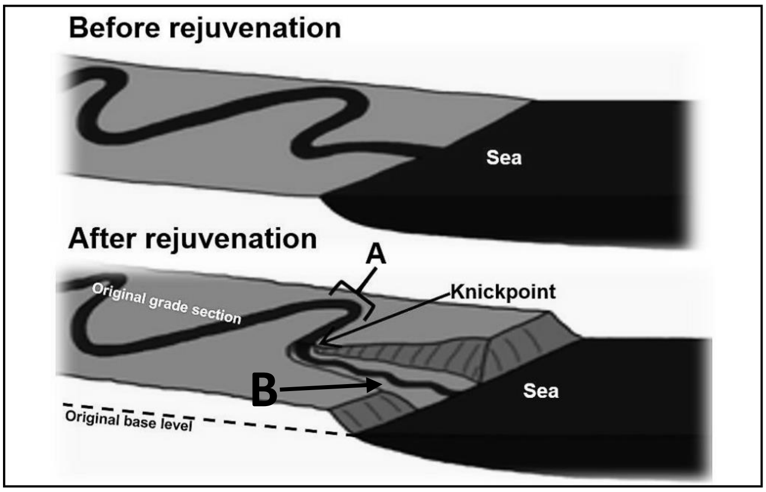
[Accept any TWO – MUST be a description]

Relief, Soil type, Rock type, Soil moisture, Vegetation, Rainfall, Land use

[15]

QUESTION 7: Fluvial Processes

7. Study the figure below which illustrates river rejuvenation and answer the questions that follow.



7.1. Define the term *river rejuvenation*. (1 x 2) (2)

[Concept]

An increase in river energy

7.2. Name the rejuvenation feature indicated at B. (1 x 1) (1)

Incised meander (accept Valley within a Valley)

7.3. Explain how river rejuvenation would have led to the creation of the feature found at B. (1 x 2) (2)

[Accept reasonable answers]

An increase in energy would have resulted in an increase in erosion

7.4. Discuss TWO positive effects of river rejuvenation. (2 x 2) (4)

[Accept logical answers]

More water downstream

Increased potential for farming

Increased potential for tourism

Increased potential for electricity generation

More alluvium brought down (increased soil fertility)

Expansion of habitats

7.5. Draw a diagram and fully explain the formation of a knickpoint waterfall.

NOTE: 2 marks will be awarded for the diagram and 4 marks will be awarded for the discussion around the formation of said diagram. (3 x 2) (6)

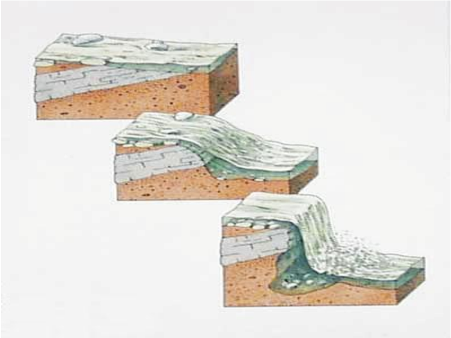


Diagram – evidence of erosion must be present. Resistant rock strata must be present

Explanation [Any TWO of the following]

These develop when water crosses a layer of resistant rock that dips (points) upstream

The softer rock beneath or downstream of this layer is eroded by the water, creating a drop

This resistant tip is termed the lip or fall-maker

Hydraulic action and abrasion creates a plunge pool at the bottom

[15]

QUESTION 8: River Management

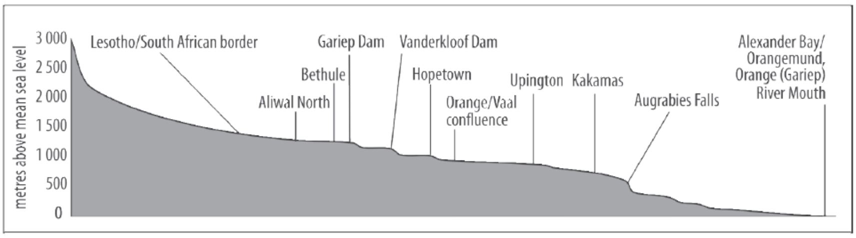
Study the article below and answer the questions that follow.

The Orange (Gariep) River: South Africa's biggest river

The Orange (Gariep) River is South Africa's main river. It rises in the Maluti Mountains of Lesotho, flows south-west through Lesotho, meanders north-west and then west across South Africa, and finally flows through parts of the Kalahari and Namib Deserts, where it then enters the Atlantic Ocean at Oranjemund. In very dry years, it does not reach the sea.

Navigation on the river is limited due to rapids, falls, and shoals (sandbanks) in braided sections of the river.

The river provides water for irrigation and municipal use, and it generates hydroelectricity. The construction of the Vanderkloof Dam has made it possible to turn thousands of hectares of arid land into productive land. Via transfer schemes, tunnels divert water to the Great Fish and Sundays Rivers. The river mouth contains rich alluvial diamond deposits and forms a delta-type wetland.



8.1. Which large river joins the Orange River basin between Hopetown and Upington? (1 x 1) (1)

Vaal river

8.2. Give the term for a large river that flows through a desert – like the Orange (Gariep) River does. (1 x 1) (1)

Exotic river

8.3. Is the river profile shown above a transverse or longitudinal profile? (1 x 1) (1)

Longitudinal

8.4. On the river's profile, identify:

a) A knickpoint (1)

Augrabies Falls

b) The permanent base level (1)

River mouth / Ocean

c) TWO temporary base levels (2 x 1) (2)

Gariep Dam, Vanderkloof Dam

8.5. Describe TWO ways in which the Orange (Gariep) River is under pressure (indirectly referred to in the extract), and suggest TWO management strategies or solutions for these problems. (4 x 2) (8)

[Accept reasonable answers]
Ways the river is under pressure
Climate change, drought, damming and irrigation schemes

Management strategies
Creating legislation that will control construction
Deforestation must be controlled
EIAs are essential
Wetlands must be protected
Clearing of alien plants must be controlled
Construction must be carefully planned
Farmers and the public need to be educated on the importance of such systems

[15]

TOTAL SECTION B: [60]

SECTION C - Mapwork

RESOURCE MATERIAL

1. 2930CA MERRIVALE and an orthophoto 2930CA 5 MERRIVALE

GENERAL INFORMATION ON MERRIVALE

Merrivale rests not far from the banks of the Howick Dam.

The Main road and the Umgeni River act as dividers. But tacked on to the southern tip of Howick as it is, Merrivale might have its own train station and accompanying old-style character railway houses, but today it merges seamlessly with the larger Howick

Just alongside Merrivale is the Sakabula Golf and Country Estate, a major draw card to the area for golfers – the only par 73 rated golf course in the province with two very different nine hole courses, set in a rural environment.

There are also a series of falls in the area. Howick Falls, after which the town is named, rush 95m down dolerite cliffs on the Umgeni river, whilst Cascade Falls, Shelter Falls and Karkloof Falls all lie within the vicinity of Howick, and make a good day trip.

QUESTION 9: Calculations

9.1. The altitude of the communication tower located in block **C3** on the topographic map is approximately … metres. NOTE: Please round to the nearest 20m. (1 x 1) (1)

1100m

9.2. Would you be able to see the cemetery (block **D3**) from the rifle range (block **C3**) if you had the most powerful binoculars known to man? (1 x 1) (1)

No.

9.3. Give a reason for your answer to QUESTION 9.2. (1 x 2) (2)

There is a mountain in the way.

9.4. Choose the correct option below.

The area covered by the orthophoto map as represented on the topographic map is approximately:

B. 2.775km².

(1 x 2) (2)

9.5. Calculate the current magnetic declination for 2023. Show ALL calculations. Use the following headings as a guide:

The difference in years: 2023 – 2016 = 7years

Mean annual change: 9'

Total change: 7 x 9 = 63'

Magnetic declination for 2023: 25°45' West

(4 x 1) (4)

[10]

QUESTION 10: Interpretation

10.1. Refer to the river system **I** in block **D3** and **C3**.

10.1.1. Identify the drainage pattern of the river system in block **D3** and **C3**. (1 x 1) (1)

Dendritic

10.1.2. In what general direction is this river flowing in? (1 x 1) (1)

North (accept NW)

10.1.3. Provide map evidence for the answer in QUESTION 10.1.2. (1 x 2) (2)

Contours decrease in that direction. Tributaries join at acute angles pointing North

10.1.4. How can people make use the dams in block **A4**? (1 x 2) (2)

[Accept logical answers]

Hydroelectricity, farming irrigation, water supply for people etc.

10.2. Consider the Orthophoto map.

10.2.1. What feature is found between **6** and **7** on the Orthophoto map? (1 x 1) (1)

Power lines

10.2.2. Based on the shadows of the trees in block **B3**, was this Orthophoto likely taken in the morning or the afternoon? (1 x 1) (1)

Morning

10.2.3. Describe the purpose of the rows of trees in blocks **B4** and **A4**. (1 x 2) (2)

Demarking boundaries

10.2.4. "Pollution domes occur very rarely in this area". Provide ONE reason from the orthophoto map to justify this statement. (1 x 2) (2)

Tons of vegetation, very little urban development, very little industry

[12]

QUESTION 11: GIS

1.1. Is feature **F** on the topographic map an example of a point or polygon feature? (1 x 1) (1)

Polygon

1.2. Which format, raster or vector, would be best for showing continuous categories of data such as temperature? (1 x 1) (1)

Raster

1.3. Name the GIS concept that involves combining data from different systems to produce new information. (1 x 1) (1)

Integration [Accept other reasonable terms]

1.4. Name ONE layer that would have likely been considered by GIS consultants before developing the landing strip in block **C3**. (1 x 1) (1)

[Accept reasonable responses]

Gradient, vegetation, settlement, biodiversity etc.

1.5. The main road that crosses the dam in block **B3** required very careful and thought-out planning. Explain how GIS would have been used in the construction of this road over this dam wall. (2 x 2) (4)

[Accept logical answers]

GIS layering would have been used to understand geology.

GIS would have been used to understand climate and weather (also could link to flooding risks)

GIS would have been used to assess the need for such construction (e.g. reducing flooding or providing more water to industry/farming)

Etc.

[8]

TOTAL SECTION C: [30]

GRAND TOTAL: 150 Marks